Fully Coupled 1-D Schrodinger–Poisson Solver for a (100) Si MOS Inversion Layer

Including $p(\phi)$, $N_A^-(\phi)$, and a Three-Subband Reduced Model

Numerical derivation and implementation note

Abstract

This note replaces the earlier strong-inversion approximation $p \approx 0$ by a nonlinear Poisson equation that retains holes, incomplete acceptor ionization, and the quantum electron density. The quantum density is obtained from two (100)-Si valley families with Fermi–Dirac subband occupation. The electrostatic potential is determined by a MOS oxide/gate Robin boundary condition at the interface and a bulk Dirichlet condition. Python and MATLAB implementations are included. The numerical example follows the benchmark conditions used by Janik and Majkusiak: $N_A = 10^{17}\,\mathrm{cm}^{-3}$, $t_{ox} = 3\,\mathrm{nm}$, $T = 300\,\mathrm{K}$, and $V_G = 4\,\mathrm{V}$, while the flat-band voltage remains an explicit model parameter because the paper does not fully specify the absolute gate/oxide/Si band alignment.

Nomenclature

x	Distance normal to the Si/SiO_2 interface	g_v	Valley degeneracy
ϕ	Electrostatic potential in silicon	$m^*_{z,v}$	Confinement effective mass
ϕ_s	Surface electrostatic potential, $\phi(0)$	$m^*_{d,v}$	In-plane density-of-states mass
E_c, E_v	Conduction- and valence-band edges	m_l, m_t	Longitudinal and transverse Si masses
E_F	Equilibrium Fermi level	q	Magnitude of the electronic charge
E_{vj}	Energy of subband j in valley family v	k_B	Boltzmann constant
ψ_{vj}	Normalized envelope wavefunction	T	Absolute temperature
n_{QM}	Quantum electron concentration	V_T	Thermal voltage, k_BT/q
p	Hole concentration	ε_{Si}	Silicon permittivity
N_A	Total acceptor concentration	ε_{ox}	Oxide permittivity
N_A^-	Ionized acceptor concentration	C_{ox}	Oxide capacitance per unit area
N_{vj}	Sheet population of subband (v, j)	t_{ox}	Oxide thickness
N_s	Number of retained quantum subbands per valley family	V_G	Applied gate voltage
		V_{FB}	Flat-band voltage
$n_3(x)$	Electron density from the three-subband reduced model	L	Silicon simulation-domain depth
		Δx	Spatial mesh spacing
$n_{10}(x)$	Ten-state-per-valley numerical reference density	H_v	Finite-difference Hamiltonian for valley v
		R_i	Nonlinear Poisson residual at node i

1 Physical model

For a planar MOS structure, confinement is along x, normal to the Si/SiO_2 interface. The electron envelope function in valley family v and subband j obeys

$$-\frac{\hbar^2}{2m^*_{z,v}}\frac{d^2\psi_{vj}}{dx^2} + E_c(x)\psi_{vj}(x) = E_{vj}\psi_{vj}(x), \tag{1}$$

with $E_c(x) - E_{c,\mathrm{bulk}} = -q\phi(x)$. For (100) Si there are two valley families:

Family	Degeneracy g_v	confinement mass $m_{z,v}$	in-plane DOS mass $m_{d,v}$
L	2	m_l	m_t
T	4	m_t	$\sqrt{m_l m_t}$

The numerical values used here are $m_l = 0.9163 m_0$ and $m_t = 0.2245 m_0$.

1.1 Fermi–Dirac subband population

The sheet population of subband (v, j) is

$$N_{vj} = \frac{g_v m^*_{d,v} k_B T}{\pi \hbar^2} \ln\left[1 + \exp\left(\frac{E_F - E_{vj}}{k_B T}\right)\right]. \tag{2}$$

The quantum electron concentration is therefore

$$\boxed{n_{\mathrm{QM}}(x) = \sum_{v,j} N_{vj} |\psi_{vj}(x)|^2 + n_0.} \tag{3}$$

The small n_0 term restores the correct equilibrium minority-electron floor in the far bulk.

2 Three-subband reduced model

A reduced model is obtained by retaining only the lowest three confined states in each valley family,

$$\boxed{n_3(x) = \sum_{v \in \{L,T\}} \sum_{j=1}^{3} N_{vj} |\psi_{vj}(x)|^2 + n_0.} \tag{4}$$

The reference calculation used in this report retains ten finite-box eigenstates per valley family,

$$n_{10}(x) = \sum_{v \in \{L,T\}} \sum_{j=1}^{10} N_{vj} |\psi_{vj}(x)|^2 + n_0. \tag{5}$$

Both models are solved *self-consistently*: changing the number of retained subbands changes the electron density, which changes Poisson's equation, which in turn changes the potential and the eigenstates. Thus the comparison is not a post-processing truncation of one fixed potential.

Spinelli, Benvenuti, and Pacelli used a related computational reduction: three low subbands were treated quantum mechanically and higher-energy states were represented by a classical tail. The present "3-subband" comparison intentionally isolates the effect of truncating the discrete spectrum; it does not add their separate classical-tail correction. This distinction is important when interpreting the numerical error.

Define the sheet inversion population of a model M as

$$N^{(M)}_{\mathrm{inv}} = \int_0^L [n_M(x) - n_0]\, dx, \tag{6}$$

and the local density error of the three-subband model as

$$\delta_n(x) = 100\, \frac{n_3(x) - n_{10}(x)}{n_{10}(x)} \quad [\%]. \tag{7}$$

3 Full nonlinear Poisson equation

The earlier $p \approx 0$ approximation is removed. The charge density is

$$\rho(x) = q\left[p(x) - n_{\mathrm{QM}}(x) - N_A^-(x)\right], \tag{8}$$

so that

$$\boxed{\frac{d^2\phi}{dx^2} = \frac{q}{\varepsilon_{Si}}\left[n_{\mathrm{QM}}(x) + N_A^-(\phi) - p(\phi)\right].} \tag{9}$$

3.1 Hole concentration

With nondegenerate holes,

$$p(\phi) = p_0 \exp\!\left(-\frac{\phi}{V_T}\right), \qquad V_T = \frac{k_B T}{q}. \tag{10}$$

3.2 Incomplete acceptor ionization

For an acceptor level $E_A = E_V + E_{A0}$ with degeneracy factor g_A,

$$\boxed{N_A^-(\phi) = \frac{N_A}{1 + g_A \exp\left[\frac{E_A(\phi) - E_F}{k_B T}\right]}.} \tag{11}$$

Since the semiconductor bands shift by $-q\phi$ while the equilibrium Fermi level is fixed,

$$E_F - E_V(x) = (E_F - E_V)_{\mathrm{bulk}} + q\phi(x). \tag{12}$$

Hence, in eV units,

$$N_A^-(\phi) = \frac{N_A}{1 + g_A \exp\left\{\frac{E_{A0} - [(E_F - E_V)_{\mathrm{bulk}} + \phi]}{V_T}\right\}}. \tag{13}$$

The implementation uses $E_{A0} = 45\,\mathrm{meV}$ and $g_A = 4$ as a boron-like incomplete-ionization model. These two quantities are an added modeling choice; they are not specified by the attached Janik–Majkusiak paper.

3.3 Bulk charge neutrality

The bulk Fermi level is obtained from

$$p_0 - n_0 - N_{A0}^- = 0, \tag{14}$$

with

$$p_0 = N_V \exp[-(E_F - E_V)/k_B T], \tag{15}$$

$$n_0 = N_C \exp[-(E_C - E_F)/k_B T]. \tag{16}$$

This ensures that the far-bulk Poisson source tends to zero.

4 MOS boundary conditions

At the Si/SiO_2 interface, Gauss' law through the oxide gives the Robin condition

$$\boxed{\left.\frac{d\phi}{dx}\right|_0 = \frac{C_{ox}}{\varepsilon_{Si}}\left[\phi_s - (V_G - V_{FB})\right],} \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}. \tag{17}$$

At the far end of the Si domain,

$$\boxed{\phi(L) = 0.} \tag{18}$$

Thus the surface potential is solved rather than imposed.

5 Finite-difference Hamiltonian

On a uniform mesh $x_i = i\Delta x$,

$$\frac{d^2\psi_i}{dx^2} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta x^2}. \tag{19}$$

For valley family v, define

$$t_v = \frac{\hbar^2}{2m^*_{z,v}\Delta x^2}. \tag{20}$$

Then the tridiagonal Hamiltonian is

$$H_v = \begin{bmatrix} U_1 + 2t_v & -t_v & 0 & \cdots \\ -t_v & U_2 + 2t_v & -t_v & \cdots \\ 0 & -t_v & U_3 + 2t_v & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \tag{21}$$

where $U_i = -\phi_i$ when energies are represented in eV relative to the bulk conduction-band edge.

6 Nonlinear Poisson Newton step

For interior node i,

$$R_i = \frac{\phi_{i-1} - 2\phi_i + \phi_{i+1}}{\Delta x^2} - \frac{q}{\varepsilon_{Si}}\left[n_i + N^-_{A,i}(\phi_i) - p_i(\phi_i)\right]. \tag{22}$$

The diagonal Jacobian contribution is

$$\frac{\partial R_i}{\partial \phi_i} = -\frac{2}{\Delta x^2} - \frac{q}{\varepsilon_{Si}}\left[\frac{dN^-_A}{d\phi} - \frac{dp}{d\phi}\right], \tag{23}$$

where

$$\frac{dp}{d\phi} = -\frac{p}{V_T}, \tag{24}$$

$$\frac{dN^-_A}{d\phi} = \frac{N_A}{V_T} f_A(1 - f_A), \qquad f_A = \frac{N^-_A}{N_A}. \tag{25}$$

For the separate classical comparison, $dn/d\phi = n/V_T$ is added to the Jacobian.

7 Self-consistency algorithm

The quantum density is a nonlocal functional of ϕ because changing the potential changes all eigenvalues and eigenfunctions. The implementation therefore uses a stable nested iteration:

1. Start from the fully nonlinear classical Poisson solution.
2. Choose $N_s = 10$ (reference) or $N_s = 3$ (reduced model), solve both valley-family Hamiltonians, and assemble $n_{N_s}(x)$.
3. Solve the nonlinear Poisson equation containing n_{QM}, $p(\phi)$, and $N_A^-(\phi)$ in the same residual.
4. Recompute the quantum density from the updated potential.
5. Mix the quantum density geometrically and repeat until the potential update is small.

No charge term is dropped. The charge mixing is only a numerical stabilization of the nonlocal Schrodinger response.

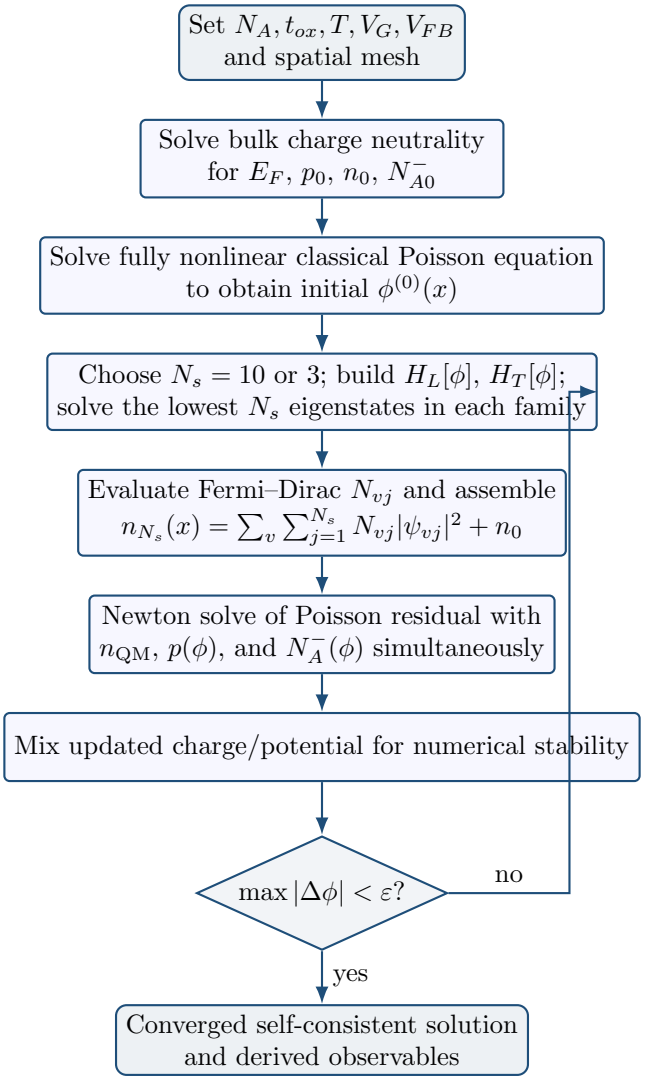


Figure 1: Algorithm flow for either the ten-state reference or the three-subband reduced model. The same nonlinear Poisson physics is used in both cases; only the number of retained eigenstates changes.

8 Numerical result

For the benchmark-style case $N_A = 10^{17}\,\mathrm{cm}^{-3}$, $t_{ox} = 3\,\mathrm{nm}$, $T = 300\,\mathrm{K}$, $V_G = 4\,\mathrm{V}$, and the exposed parameter $V_{FB} = -0.95\,\mathrm{V}$, the Python calculation converged to a quantum surface potential near $1.468\,\mathrm{V}$. The quantum electron-density maximum occurs about $0.70\,\mathrm{nm}$ from the interface. The exact absolute values depend on the chosen flat-band alignment and ionization model, but the displaced quantum peak is the same qualitative behavior shown in Janik and Majkusiak's Fig. 3.

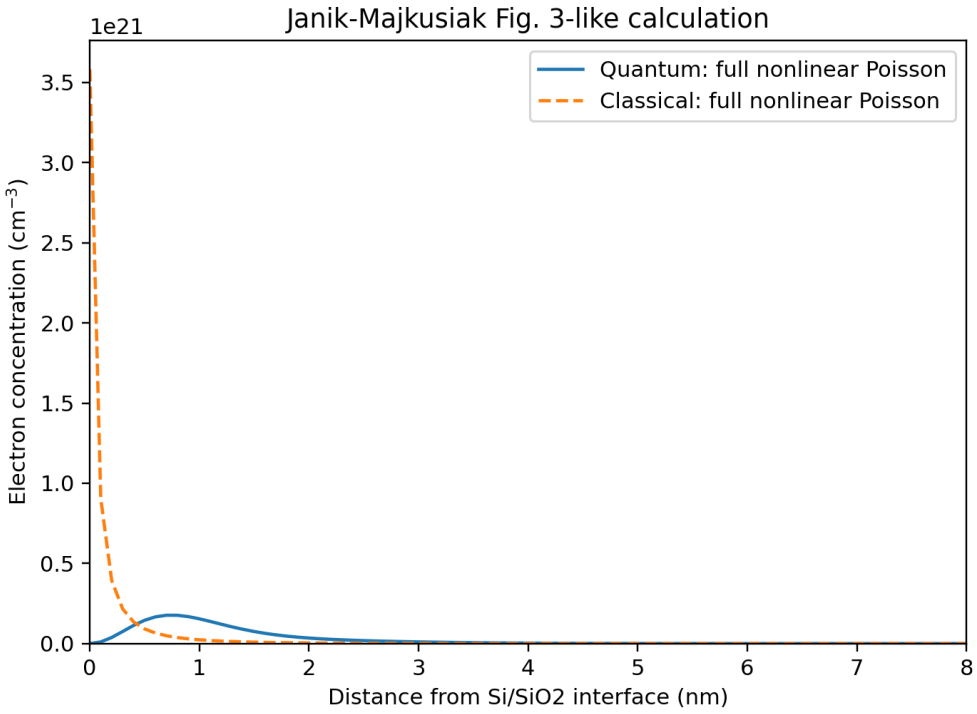


Figure 2: Fully nonlinear classical and quantum electron distributions.

9 Three-subband comparison

For the same benchmark case, the ten-state reference and the three-subband model converge to essentially the same electrostatics. The calculated values are

Model	ϕ_s (V)	$x_{\max}$ (nm)	N_{inv} (cm^{-2})
10 states/valley	1.467962	0.700	2.2537×10^{13}
3 subbands/valley	1.467962	0.700	2.2537×10^{13}

The integrated sheet-density difference is approximately

$$\boxed{\frac{N_{\mathrm{inv}}^{(3)} - N_{\mathrm{inv}}^{(10)}}{N_{\mathrm{inv}}^{(10)}} \times 100\% \simeq -0.0022\%.} \tag{26}$$

For the converged ten-state potential, the first three longitudinal-family states carry about 99.968% of that family's computed subband sheet population, and the first three transverse-family states carry about 99.9992%. Therefore, under this deep-inversion benchmark, higher discrete subbands contribute

negligibly to the total inversion population. This numerical result is specific to the stated bias, doping, domain, and band-alignment assumptions; lower transverse fields or different devices can increase the importance of higher-energy states.

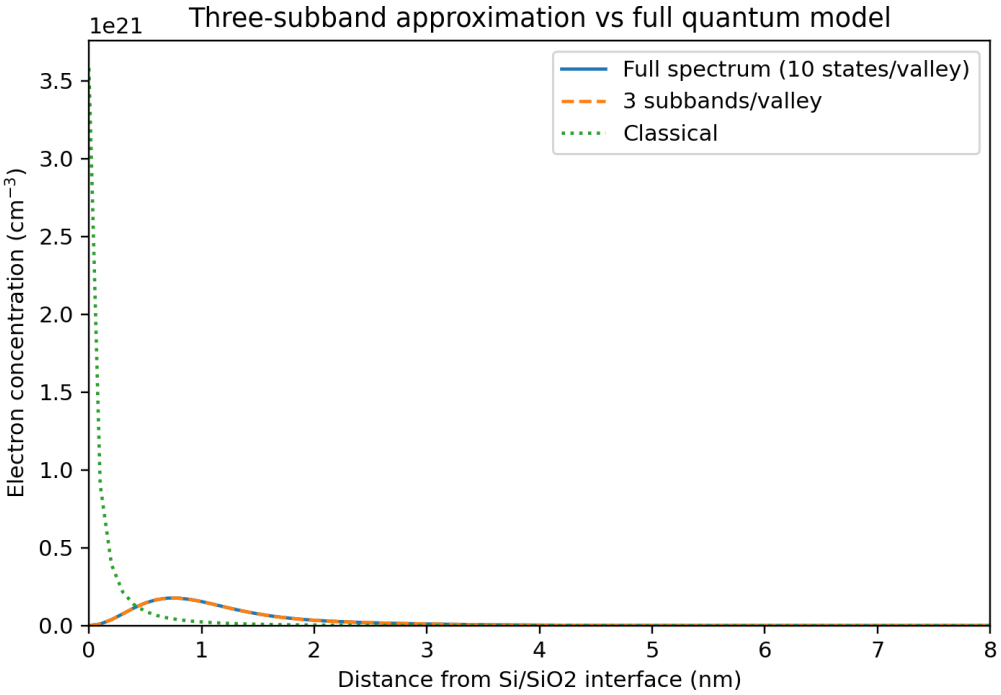


Figure 3: Self-consistent comparison of the ten-state-per-valley reference, the three-subband-per-valley reduced model, and the classical solution. The two quantum curves are nearly indistinguishable on the concentration scale.

10 Relation to the attached paper

Janik and Majkusiak formulate the electrostatics of the two-dimensional electron gas through a self-consistent one-dimensional Schrodinger–Poisson system, solve separate valley families for (100) Si, and compare quantum and classical electron profiles. Their Fig. 3 shows the classical maximum at the interface and the quantum maximum shifted into Si. They also state that rigorous numerical Schrodinger–Poisson analysis is required for accurate surface-potential calculations in ultrathin-oxide MOS structures. The present implementation extends the earlier code by retaining holes and incomplete acceptor ionization throughout the nonlinear Poisson solve. For the reduced-order comparison, it also tests an explicit three-subband truncation. Spinelli et al. independently motivate a three-subband quantum treatment supplemented by a classical high-energy tail; our comparison separates the discrete-state truncation error from that additional tail approximation.

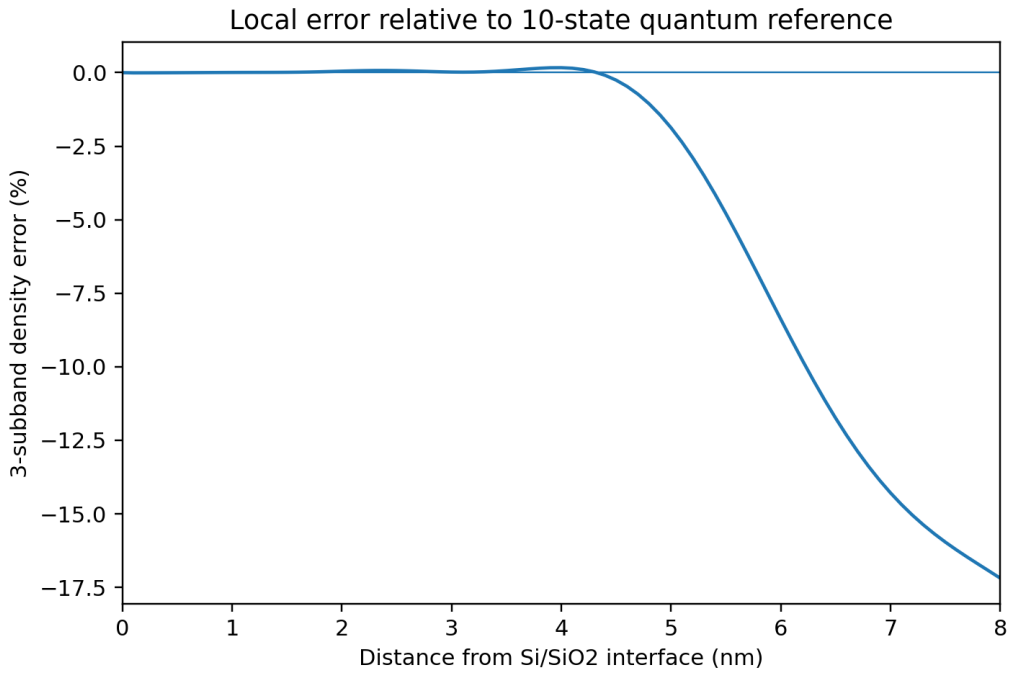


Figure 4: Local electron-density error of the three-subband model relative to the ten-state quantum reference. The relative error can become visually large only where the absolute electron density is already very small; the integrated sheet-charge error remains about -0.0022%.

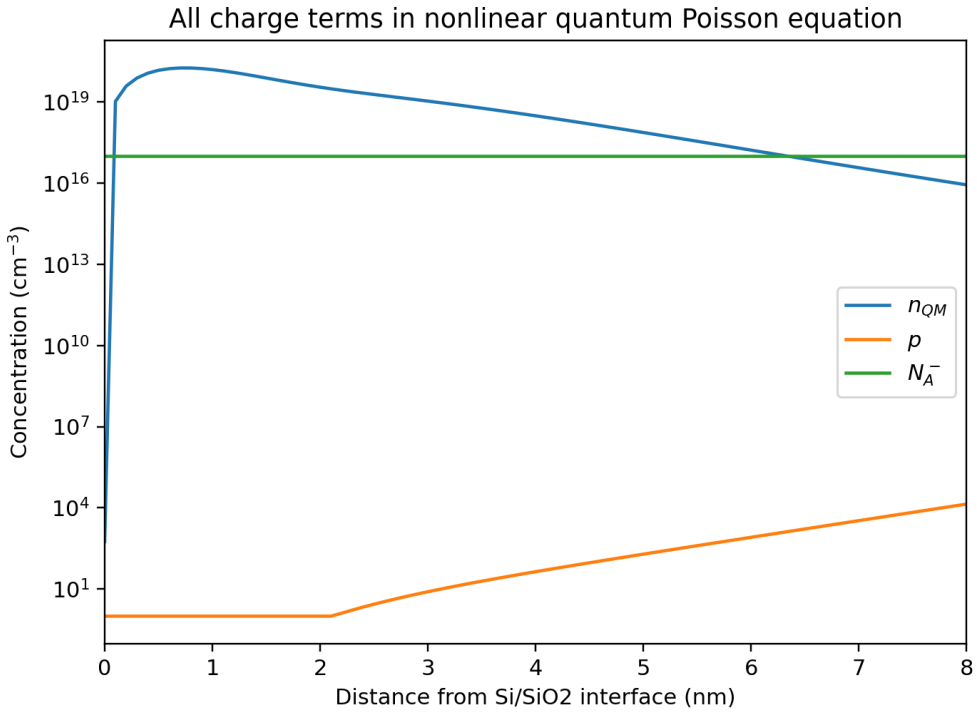


Figure 5: Quantum Poisson charge terms n_{QM}, p, and N_A^-. Holes are strongly suppressed in inversion but are retained rather than set to zero.

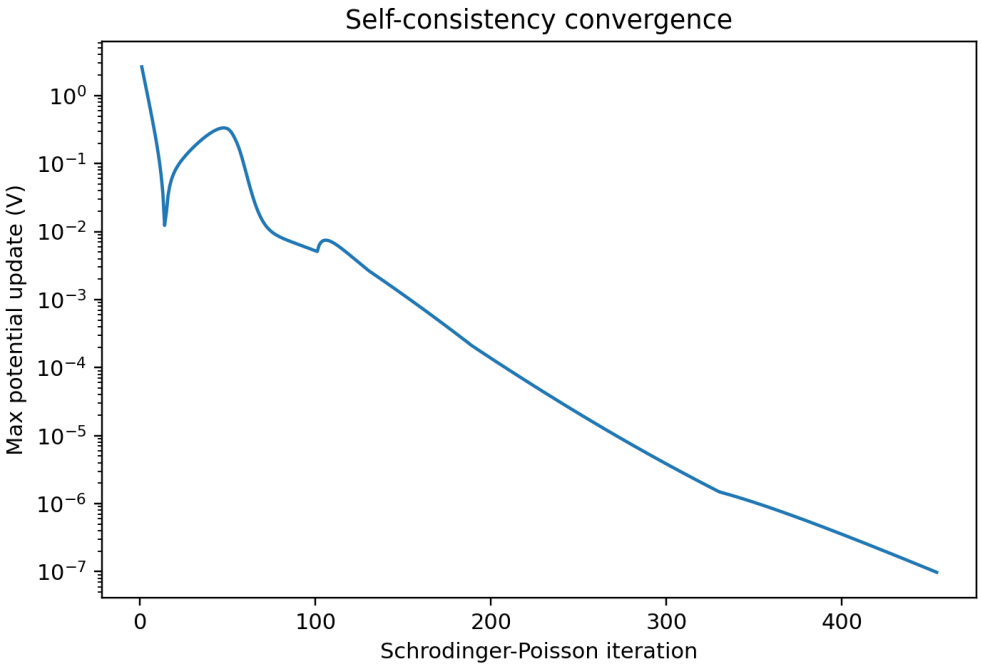


Figure 6: Outer Schrodinger–Poisson convergence history.

A Python implementation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import eigh_tridiagonal
from scipy.integrate import solve_bvp
from scipy.optimize import brentq
from scipy.sparse import lil_matrix
from scipy.sparse.linalg import spsolve

# Fully coupled 1-D MOS Schrodinger-Poisson solver
# (100) Si, two valley families, Fermi-Dirac subband occupation,
# holes and incomplete acceptor ionization retained in Poisson equation.

q = 1.602176634e-19
hbar = 1.054571817e-34
kB = 1.380649e-23
m0 = 9.1093837015e-31
eps0 = 8.8541878128e-12

T = 300.0
VT = kB*T/q
eps_si = 11.7*eps0
eps_ox = 3.9*eps0
Eg = 1.122 # eV
NA = 1.0e17*1e6 # m^-3
tox = 3.0e-9 # m
Vg = 4.0 # V
Vfb = -0.95 # V, exposed alignment parameter

# (100) silicon effective masses used in Janik-Majkusiak paper
ml = 0.9163*m0
mt = 0.2245*m0
# name, valley degeneracy, confinement mass, in-plane DOS mass
valleys = [('L', 2, ml, mt), ('T', 4, mt, np.sqrt(ml*mt))]

# Bulk DOS values. Nc from ellipsoidal valley masses; Nv conventional 300 K value.
md3 = (ml*mt*mt)**(1/3)
Nc = 2*6*(md3*kB*T/(2*np.pi*hbar**2))**1.5
Nv = 1.04e19*1e6
ni = np.sqrt(Nc*Nv)*np.exp(-Eg/(2*VT))

# Boron-like acceptor model: acceptor level Ea = Ev + EA_ION.
EA_ION = 0.045 # eV
GA = 4.0 # acceptor degeneracy factor

# Determine equilibrium bulk Fermi level with incomplete ionization.
# eta_v = (Ef-Ev)/kT in volts/eV divided by VT.
def bulk_neutrality(EfEv):
    p = Nv*np.exp(-EfEv/VT)
    n = Nc*np.exp(-(Eg-EfEv)/VT)
    NAminus = NA/(1.0 + GA*np.exp((EA_ION-EfEv)/VT))
    return p - n - NAminus

EfEv = brentq(bulk_neutrality, 0.001, 0.5)
EcEf = Eg - EfEv
p0 = Nv*np.exp(-EfEv/VT)
n0 = Nc*np.exp(-EcEf/VT)
NA0minus = NA/(1.0 + GA*np.exp((EA_ION-EfEv)/VT))

# Numerical mesh
L = 30e-9
N = 301
x = np.linspace(0.0, L, N)
dx = x[1]-x[0]
Ni = N-2
alpha = eps_ox/(eps_si*tox)
```

```
def log1pexp(a):
    a = np.asarray(a, dtype=float)
    out = np.empty_like(a)
    hi = a > 40.0
    lo = a < -40.0
    mid = ~(hi | lo)
    out[hi] = a[hi]
    out[lo] = np.exp(a[lo])
    out[mid] = np.log1p(np.exp(a[mid]))
    return out


def holes(phi):
    """Nondegenerate hole density p(phi), exact bulk value at phi=0."""
    return p0*np.exp(np.clip(-phi/VT, -100.0, 100.0))


def ionized_acceptors(phi):
    """Incomplete ionization N_A^-(phi).

    Electrostatic potential raises Ef-Ev locally by +phi (eV), since both
    semiconductor band edges shift by -q phi while Ef is fixed at equilibrium.
    """
    local_EfEv = EfEv + phi
    arg = np.clip((EA_ION-local_EfEv)/VT, -100.0, 100.0)
    return NA/(1.0 + GA*np.exp(arg))


def classical_electrons(phi):
    """Classical MB electron density for the separate reference solution."""
    return n0*np.exp(np.clip(phi/VT, -100.0, 80.0))


def schrodinger_density(phi, nstates=10):
    """Quantum electron density from two Si valley families.

    Energies are measured relative to the bulk conduction-band edge Ec_bulk.
    U(x) = Ec(x)-Ec_bulk = -phi(x) in eV.
    """
    U = -phi
    ntot = np.zeros_like(x)
    info = []
    for name, g, mz, md in valleys:
        t = (hbar*hbar/(2.0*mz*dx*dx))/q # eV
        diag = U[1:-1] + 2.0*t
        off = -t*np.ones(Ni-1)
        ns = min(nstates, Ni)
        vals, vecs = eigh_tridiagonal(diag, off, select='i', select_range=(0, ns-1))
        sheets = []
        psis = []
        for E, v in zip(vals, vecs.T):
            psi = np.zeros(N)
            psi[1:-1] = v
            psi /= np.sqrt(np.trapezoid(psi*psi, x))
            eta = ((-EcEf)-E)/VT # (Ef - Esub)/kT
            Ns = g*md*kB*T/(np.pi*hbar*hbar)*log1pexp([eta])[0]
            ntot += Ns*psi*psi
            sheets.append(Ns)
            psis.append(psi)
        info.append({'name': name, 'g': g, 'mz': mz, 'md': md,
                     'E': np.array(vals), 'Ns': np.array(sheets),
                     'psi': np.array(psis)})

    # Add the equilibrium minority density as a smooth far-bulk floor.
    ntot += n0
    return ntot, info


def gate_bc(ya, yb):
    # epsilon_si phi'(0) = Cox[phi_s-(Vg-Vfb)] ; phi(L)=0
```

```
    return np.array([ya[1] - alpha*(ya[0]-(Vg-Vfb)), yb[0]])


def solve_poisson_nonlinear(n_fixed, guess, tol=1e-7, maxit=100):
    """Damped-Newton finite-difference Poisson solve.

    Residual contains all local charge terms simultaneously:
      phi'' - q/eps_si [n_fixed + N_A^-(phi) - p(phi)] = 0.
    The quantum density is nonlocal in phi and is refreshed by the outer
    Schrodinger-Poisson iteration.
    """
    M = N-1 # phi[0:M] unknown, phi[N-1]=0 bulk Dirichlet
    ph = guess[:M].copy()
    Vgb = Vg - Vfb
    for _ in range(maxit):
        full = np.zeros(N); full[:M] = ph
        pp = holes(full)[:M]
        naa = ionized_acceptors(full)[:M]
        # derivatives wrt phi
        dpp = -pp/VT
        f = naa/NA
        dna = NA*f*(1.0-f)/VT

        R = np.zeros(M)
        J = lil_matrix((M,M))
        # second-order one-sided gate derivative
        R[0] = (-3*ph[0] + 4*ph[1] - ph[2])/(2*dx) - alpha*(ph[0]-Vgb)
        J[0,0] = -3/(2*dx) - alpha
        J[0,1] = 4/(2*dx)
        J[0,2] = -1/(2*dx)
        for i in range(1,M):
            pim1 = ph[i-1]
            pi = ph[i]
            pip1 = ph[i+1] if i+1<M else 0.0
            R[i] = (pim1 - 2*pi + pip1)/dx**2 - q/eps_si*(n_fixed[i] + naa[i] - pp[i])
            J[i,i-1] = 1/dx**2
            J[i,i] = -2/dx**2 - q/eps_si*(dna[i] - dpp[i])
            if i+1<M: J[i,i+1] = 1/dx**2
        if np.max(np.abs(R[1:])) < 1e10 and abs(R[0]) < 1e3:
            break
        delta = spsolve(J.tocsr(), -R)
        # line search on scaled update; restrict excessively large Newton moves
        maxdu = np.max(np.abs(delta))
        if maxdu > 0.15: delta *= 0.15/maxdu
        lam = 1.0
        norm0 = np.linalg.norm(R)
        accepted = False
        for _ls in range(12):
            trial = ph + lam*delta
            tfull=np.zeros(N); tfull[:M]=trial
            tpp=holes(tfull)[:M]; tna=ionized_acceptors(tfull)[:M]
            Rt=np.zeros(M)
            Rt[0]=(-3*trial[0]+4*trial[1]-trial[2])/(2*dx)-alpha*(trial[0]-Vgb)
            for i in range(1,M):
                pip1=trial[i+1] if i+1<M else 0.0
                Rt[i]=(trial[i-1]-2*trial[i]+pip1)/dx**2-q/eps_si*(n_fixed[i]+tna[i]-tpp[i])
            if np.linalg.norm(Rt) < norm0:
                ph=trial; accepted=True; break
            lam*=0.5
        if not accepted:
            ph += 0.05*delta
        if np.max(np.abs(lam*delta)) < tol:
            break
    out=np.zeros(N); out[:M]=ph
    return out


def solve_classical(guess=None, tol=1e-7, maxit=150):
    if guess is None:
        guess = 0.9*np.exp(-x/5e-9)
```

```
    M=N-1; ph=guess[:M].copy(); Vgb=Vg-Vfb
    for _ in range(maxit):
        full=np.zeros(N); full[:M]=ph
        nn=classical_electrons(full)[:M]
        pp=holes(full)[:M]
        naa=ionized_acceptors(full)[:M]
        dnn=nn/VT; dpp=-pp/VT
        f=naa/NA; dna=NA*f*(1-f)/VT
        R=np.zeros(M); J=lil_matrix((M,M))
        R[0]=(-3*ph[0]+4*ph[1]-ph[2])/(2*dx)-alpha*(ph[0]-Vgb)
        J[0,0]=-3/(2*dx)-alpha; J[0,1]=4/(2*dx); J[0,2]=-1/(2*dx)
        for i in range(1,M):
            pip1=ph[i+1] if i+1<M else 0.0
            R[i]=(ph[i-1]-2*ph[i]+pip1)/dx**2-q/eps_si*(nn[i]+naa[i]-pp[i])
            J[i,i-1]=1/dx**2
            J[i,i]=-2/dx**2-q/eps_si*(dnn[i]+dna[i]-dpp[i])
            if i+1<M: J[i,i+1]=1/dx**2
        delta=spsolve(J.tocsr(),-R)
        maxdu=np.max(np.abs(delta))
        if maxdu>0.10: delta*=0.10/maxdu
        ph += 0.5*delta
        if np.max(np.abs(0.5*delta))<tol: break
    out=np.zeros(N); out[:M]=ph
    return out


# Classical reference
phi_c = solve_classical()
nc = classical_electrons(phi_c)
pc = holes(phi_c)
NAc = ionized_acceptors(phi_c)


def solve_quantum_model(nstates, phi_init=None, max_outer=600):
    """Self-consistent quantum solution retaining nstates per valley family."""
    phi_q = phi_c.copy() if phi_init is None else phi_init.copy()
    n_work, info = schrodinger_density(phi_q, nstates=nstates)
    history = []
    for it in range(max_outer):
        phi_raw = solve_poisson_nonlinear(n_work, phi_q, tol=1e-7)
        nq_new, info = schrodinger_density(phi_raw, nstates=nstates)
        err = np.max(np.abs(phi_raw-phi_q))
        history.append(err)
        beta = 0.01 if it < 100 else 0.02
        floor = max(n0, 1.0)
        logn = (1-beta)*np.log(np.maximum(n_work, floor)) + beta*np.log(np.maximum(nq_new, floor))
        n_work = np.exp(logn)
        phi_q = 0.25*phi_raw + 0.75*phi_q
        dens_err = np.max(np.abs(np.log(np.maximum(nq_new,floor)/np.maximum(n_work,floor))))
        if err < 2e-5 and dens_err < 2e-3:
            break
    nq, info = schrodinger_density(phi_q, nstates=nstates)
    return phi_q, nq, info, np.asarray(history)


# "Full" numerical reference: ten finite-box eigenstates per valley family.
phi_full, nq_full, info_full, hist_full = solve_quantum_model(10)
# Reduced model: only the lowest three subbands in each valley family.
phi_3, nq_3, info_3, hist_3 = solve_quantum_model(3, phi_init=phi_full)

pq_full = holes(phi_full); NAq_full = ionized_acceptors(phi_full)
pq_3 = holes(phi_3); NAq_3 = ionized_acceptors(phi_3)

# Diagnostics and integrated sheet charge.
def diagnostics(phi, n):
    xmax = x[np.argmax(n)]
    xav = np.trapezoid(x*n, x)/np.trapezoid(n, x)
    Ns = np.trapezoid(np.maximum(n-n0,0.0), x)
    return xmax, xav, Ns
```

```
xmax_f,xav_f,Ns_f=diagnostics(phi_full,nq_full)
xmax_3,xav_3,Ns_3=diagnostics(phi_3,nq_3)
rel_Ns=(Ns_3-Ns_f)/Ns_f*100.0
surf_rel=(phi_3[0]-phi_full[0])*1e3

print(f'Bulk Ef-Ev = {EfEv:.6f} eV, Ec-Ef = {EcEf:.6f} eV')
print(f'Full model: phi_s={phi_full[0]:.6f} V, xmax={xmax_f*1e9:.3f} nm, xav={xav_f*1e9:.3f} nm, Ns={Ns_f/1e4:.4e}
    cm^-2')
print(f'3-subband : phi_s={phi_3[0]:.6f} V, xmax={xmax_3*1e9:.3f} nm, xav={xav_3*1e9:.3f} nm, Ns={Ns_3/1e4:.4e} cm
    ^-2')
print(f'3-subband vs full: Delta phi_s={surf_rel:.3f} mV, Delta Ns={rel_Ns:.4f}%')
for tag,info in [('full',info_full),('3sub',info_3)]:
   for a in info:
      print(tag,a['name'],'E [eV rel Ec_bulk]=',np.round(a['E'][:6],5))

# Save comparison data.
rel_profile = 100.0*(nq_3-nq_full)/np.maximum(nq_full,1.0)
out=np.c_[x*1e9,phi_full,phi_3,phi_c,nq_full/1e6,nq_3/1e6,nc/1e6,rel_profile]
np.savetxt('/mnt/data/janik_3subband_comparison.csv',out,delimiter=',',
 header='x_nm,phi_full_V,phi_3sub_V,phi_classical_V,n_full_cm-3,n_3sub_cm-3,n_classical_cm-3,
     relative_density_error_percent',comments='')

m=x<=8e-9
plt.figure(figsize=(6.6,4.8))
plt.plot(x[m]*1e9,nq_full[m]/1e6,label='Full spectrum (10 states/valley)')
plt.plot(x[m]*1e9,nq_3[m]/1e6,'--',label='3 subbands/valley')
plt.plot(x[m]*1e9,nc[m]/1e6,':',label='Classical')
plt.xlabel('Distance from Si/SiO2 interface (nm)')
plt.ylabel('Electron concentration (cm$^{-3}$)')
plt.title('Three-subband approximation vs full quantum model')
plt.xlim(0,8); plt.ylim(bottom=0); plt.legend(); plt.tight_layout()
plt.savefig('/mnt/data/janik_3subband_comparison.png',dpi=220); plt.close()

plt.figure(figsize=(6.6,4.5))
plt.plot(x[m]*1e9,rel_profile[m])
plt.axhline(0,linewidth=0.8)
plt.xlabel('Distance from Si/SiO2 interface (nm)')
plt.ylabel('3-subband density error (%)')
plt.title('Local error relative to 10-state quantum reference')
plt.xlim(0,8); plt.tight_layout()
plt.savefig('/mnt/data/janik_3subband_relative_error.png',dpi=220); plt.close()
```

B MATLAB implementation

```
% Fully nonlinear 1-D Schrodinger-Poisson MOS solver
% Comparison: 10-state reference vs lowest 3 subbands per Si valley family.
clear; clc;
q=1.602176634e-19; hbar=1.054571817e-34; kB=1.380649e-23;
m0=9.1093837015e-31; eps0=8.8541878128e-12;
T=300; VT=kB*T/q; eps_si=11.7*eps0; eps_ox=3.9*eps0;
Eg=1.122; NA=1e17*1e6; tox=3e-9; Vg=4.0; Vfb=-0.95;
ml=0.9163*m0; mt=0.2245*m0; Nv=1.04e19*1e6;
md3=(ml*mt*mt)^(1/3); Nc=2*6*(md3*kB*T/(2*pi*hbar^2))^(3/2);
EA=0.045; gA=4;
fun=@(EfEv) Nv*exp(-EfEv/VT)-Nc*exp(-(Eg-EfEv)/VT)-NA/(1+gA*exp((EA-EfEv)/VT));
EfEv=fzero(fun,[0.001 0.5]); EcEf=Eg-EfEv;
p0=Nv*exp(-EfEv/VT); n0=Nc*exp(-EcEf/VT);
L=30e-9; N=301; x=linspace(0,L,N)'; dx=x(2)-x(1);
alpha=eps_ox/(eps_si*tox);
holes=@(phi) p0.*exp(max(min(-phi/VT,100),-100));
NAminus=@(phi) NA./(1+gA.*exp(max(min((EA-(EfEv+phi))/VT,100),-100)));
nclass=@(phi) n0.*exp(max(min(phi/VT,80),-100));

% Fully nonlinear classical reference.
phi_c=0.9*exp(-x/5e-9);
for it=1:150
    [phi_new,du]=poisson_newton(phi_c,nclass(phi_c),true,x,dx,alpha,Vg,Vfb,q,eps_si,holes,NAminus,nclass,VT,NA);
    phi_c=phi_new; if du<1e-7, break; end
end
nc=nclass(phi_c);

% Quantum 10-state and 3-subband models.
[phi_full,nfull,hfull]=solve_quantum(phi_c,10,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0,alpha,Vg,Vfb,eps_si,holes,NAminus,
     nclass,VT,NA);
[phi_3,n3,h3]=solve_quantum(phi_full,3,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0,alpha,Vg,Vfb,eps_si,holes,NAminus,nclass,VT,
     NA);

Nsfull=trapz(x,max(nfull-n0,0)); Ns3=trapz(x,max(n3-n0,0));
[~,i1]=max(nfull); [~,i3]=max(n3);
fprintf('10-state: phi_s=%.6f V, xmax=%.3f nm, Ns=%.4e cm^-2\n',phi_full(1),x(i1)*1e9,Nsfull/1e4);
fprintf('3-subband: phi_s=%.6f V, xmax=%.3f nm, Ns=%.4e cm^-2\n',phi_3(1),x(i3)*1e9,Ns3/1e4);
fprintf('Delta Ns=%.5f %%\n',100*(Ns3-Nsfull)/Nsfull);

mask=x<=8e-9;
figure; plot(x(mask)*1e9,nfull(mask)/1e6,'LineWidth',1.5); hold on;
plot(x(mask)*1e9,n3(mask)/1e6,'--','LineWidth',1.5);
plot(x(mask)*1e9,nc(mask)/1e6,':','LineWidth',1.5);
xlabel('Distance from Si/SiO2 interface (nm)'); ylabel('Electron concentration (cm^{-3})');
legend('10-state reference','3 subbands/valley','Classical'); grid on;
writematrix([x*1e9 phi_full phi_3 phi_c nfull/1e6 n3/1e6 nc/1e6], 'janik_3subband_comparison_matlab.csv');

function [phi_q,nq,hist]=solve_quantum(phi_init,nstates,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0,alpha,Vg,Vfb,eps_si,holes,
     NAminus,nclass,VT,NA)
phi_q=phi_init; [nwork,~]=qdensity(phi_q,nstates,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0); hist=[];
for it=1:600
    [phi_raw,~]=poisson_newton(phi_q,nwork,false,x,dx,alpha,Vg,Vfb,q,eps_si,holes,NAminus,nclass,VT,NA);
    [nnew,~]=qdensity(phi_raw,nstates,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0);
    err=max(abs(phi_raw-phi_q)); hist(end+1)=err;
    beta=0.01; if it>=100, beta=0.02; end
    floorv=max(n0,1); nwork=exp((1-beta)*log(max(nwork,floorv))+beta*log(max(nnew,floorv)));
    phi_q=0.25*phi_raw+0.75*phi_q;
    if err<2e-5, break; end
end
[nq,~]=qdensity(phi_q,nstates,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0);
end

function [phi,du]=poisson_newton(guess,nfix,isclass,x,dx,alpha,Vg,Vfb,q,eps_si,holes,NAminus,nclass,VT,NA)
N=length(x); M=N-1; ph=guess(1:M); Vgb=Vg-Vfb; du=inf;
for it=1:120
 full=[ph;0]; pp=holes(full); naa=NAminus(full); if isclass, nn=nclass(full); else, nn=nfix; end
 R=zeros(M,1); J=spalloc(M,M,3*M);
```

```
 R(1)=(-3*ph(1)+4*ph(2)-ph(3))/(2*dx)-alpha*(ph(1)-Vgb);
 J(1,1)=-3/(2*dx)-alpha; J(1,2)=4/(2*dx); J(1,3)=-1/(2*dx);
 dpp=-pp/VT; frac=naa/NA; dna=NA.*frac.*(1-frac)/VT; if isclass, dnn=nn/VT; else, dnn=zeros(size(nn)); end
 for i=2:M
   pip1=0; if i<M, pip1=ph(i+1); end
   R(i)=(ph(i-1)-2*ph(i)+pip1)/dx^2-q/eps_si*(nn(i)+naa(i)-pp(i));
   J(i,i-1)=1/dx^2; J(i,i)=-2/dx^2-q/eps_si*(dnn(i)+dna(i)-dpp(i)); if i<M, J(i,i+1)=1/dx^2; end
 end
 d=J\(-R); if max(abs(d))>0.1, d=d*(0.1/max(abs(d))); end
 ph=ph+0.5*d; du=max(abs(0.5*d)); if du<1e-7, break; end
end
phi=[ph;0];
end

function [ntot,levels]=qdensity(phi,nstates,x,dx,hbar,q,kB,T,ml,mt,EcEf,n0)
N=length(x); Ni=N-2; U=-phi; ntot=zeros(N,1); V={2,ml,mt;4,mt,sqrt(ml*mt)}; levels=cell(2,1); VT=kB*T/q;
for iv=1:2
 g=V{iv,1}; mz=V{iv,2}; md=V{iv,3}; t=(hbar^2/(2*mz*dx^2))/q;
 main=U(2:end-1)+2*t; off=-t*ones(Ni-1,1); H=spdiags([off main off],[-1 0 1],Ni,Ni);
 [vec,D]=eigs(H,nstates,'smallestreal'); E=diag(D); [E,ord]=sort(E); vec=vec(:,ord);
 for j=1:nstates
   psi=zeros(N,1); psi(2:end-1)=vec(:,j); psi=psi/sqrt(trapz(x,psi.^2));
   eta=((-EcEf)-E(j))/VT; if eta>40,l=eta; elseif eta<-40,l=exp(eta); else,l=log1p(exp(eta)); end
   Ns=g*md*kB*T/(pi*hbar^2)*l; ntot=ntot+Ns*psi.^2;
 end
 levels{iv}=E;
end
ntot=ntot+n0;
end
```

C References

References

[1] T. Janik and B. Majkusiak, "Analysis of the MOS Transistor Based on the Self-Consistent Solution to the Schrodinger and Poisson Equations and on the Local Mobility Model," *IEEE Transactions on Electron Devices*, vol. 45, no. 6, pp. 1263–1271, June 1998.

[2] A. S. Spinelli, A. Benvenuti, and A. Pacelli, "Self-Consistent 2-D Model for Quantum Effects in n-MOS Transistors," *IEEE Transactions on Electron Devices*, vol. 45, no. 6, pp. 1342–1349, June 1998.